

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 39

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Correct answer only.</i> • B (1)	1
01.2	<i>1 mark for each point.</i> • Number all possible sampling sites (eg on a grid map) (1) • Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers) (1)	2
01.3	<i>1 mark for each point.</i> • Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis (1) • So the sample does not represent the whole area and conclusions are invalid (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
02.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $38.0 / 0.002$ (1) • $= 19\,000$ (19 000 times / 1.9×10^4) (1)	2
02.2	<i>1 mark.</i> • Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain) (1)	1

Question	Marking guidance	Marks
02.3	<i>1 mark for each point.</i> • Persistence: the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain (1) • Liposolubility: the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine (1) • Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level (1) • Herons have long lives so accumulate over time (1)	4
02.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification (1) • They persist for decades so the damage continued long after use (1) • Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify (1) • Benefit of preventing malaria deaths outweighs the risk (risk:benefit) (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	
03.1	<i>1 mark for each point.</i> • pH 5.6 (1) • Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid (1)	2
03.2	<i>1 mark for each point.</i> • Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) (1) • Granite does not neutralise acids so lake water in A is acidic (4.6) (1) • Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) (1) • Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food; lichens and other food organisms lost (1)	4
03.3	<i>1 mark for each point.</i> • Corrodes metals: railway lines, railings, pipes, pylons, overhead power lines (1) • Dissolves / weakens limestone and marble buildings and statues (eg York Minster) (1)	2
03.4	<i>1 mark for each point.</i> • Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide (1) • Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO₂ alone gives the weaker sulfurous acid) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	

Question	Marking guidance	Marks
04.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • High-level: used fuel rods / fission products separated in reprocessing; vitrification in molten glass in sealed stainless steel containers, surrounded by concrete, air-cooled to remove decay heat (Sellafield) (2) • Intermediate: fuel cladding / metal tubes around fuel rods, filters from reprocessing; mixed with cement (encapsulated) in stainless steel drums, stored in concrete vaults (2) • Low-level solid: contaminated clothing, equipment; sealed in plastic bags inside steel drums / containers in a concrete-lined, monitored landfill (Drigg) (2)	6
04.2	<i>1 mark for each point.</i> • Radioactive decay of the concentrated fission products generates heat which could damage containers (1) • Isotopes with long half-lives (plutonium 24 400 years) remain hazardous for thousands of years (1)	2
04.3	<i>1 mark for each point.</i> • Liquids: filtered and passed through ion-exchange (sands / resins) to remove radionuclides, then discharged (monitored by CPA / CGM) (1) • Gases: filtered (HEPA / charcoal filters) to remove particles and iodine before release (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 303; AQA spec 3.4.3.2.14.	
05.1	<i>1 mark for each point.</i> • Molten iron from the blast furnace is converted directly to steel on the same site (1) • So it does not cool and need re-melting; transport energy between sites is also avoided; waste heat from one process used by another (eg Kalundborg) (1)	2
05.2	<i>1 mark for each point.</i> • Collecting and processing small dispersed quantities can use more energy in transport than is saved (1) • Mass production and large-scale mining have economies of scale; some materials (glass) take more energy to make than plastic but can be refilled many times (1)	2
05.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Transmitting at high voltage means low current so less energy is lost as heat in the cables (loss proportional to current squared); overhead cables cheaper than underground which need cooling (2) • ICT / smart grids co-ordinate real-time data on demand and supply from many sources, switching generation, storage (pumped storage, batteries) and flexible loads so surplus is not wasted and inefficient peaking plants are avoided (2)	4
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Proximity to the grid / demand to reduce transmission losses and cable costs (1) • Access to cooling water for thermal stations (1) • Resource availability (wind, tidal range, fuel deliveries) (1) • Land availability and planning / public acceptance; stable geology (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 234-235; AQA spec 3.3.4.2.3.	

Question	Marking guidance	Marks
06.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(68\,000 - 7\,000) / 68\,000 \times 100$ (1) • = 89.7% (accept 90%) (1)	2
06.2	<i>1 mark for each point.</i> • Inflow from the rivers was reduced (sometimes to zero) so less water entered while evaporation continued (1) • Evaporation removes pure water leaving the dissolved salts behind, so they became concentrated in a smaller volume (1)	2
06.3	<i>1 mark for each point.</i> • Salinity rose above the range of tolerance of the freshwater fish species / osmotic stress; pollution by pesticides, fertilisers and industrial waste concentrated in the lake (1) • Exposed lake bed of salt and pesticide-contaminated dust blown by wind onto farmland and settlements (health problems, soil salinisation) (1) • Loss of wetland habitats and wildlife; local climate more extreme (loss of the moderating effect of the water body) (1) • Loss of livelihoods; conflict between the countries sharing the rivers (1)	4
06.4	<i>1 mark for each point.</i> • Upstream abstraction for irrigation reduces the flow available downstream (1) • Upstream HEP dams hold water in summer for winter generation, reducing summer irrigation water downstream; growing populations increase demand (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117-118; AQA spec 3.2.2.1-3.2.2.2.	
07.1	<i>1 mark for each point.</i> • 2020 has higher peak flow in the wet season (32 vs $22\text{ m}^3\text{ s}^{-1}$ in March) (1) • 2020 has much lower dry-season flow (2 vs $9\text{ m}^3\text{ s}^{-1}$ in July / August) (1) • Flow is much more variable / seasonal in 2020; the river almost dries up (1)	3
07.2	<i>1 mark for each point.</i> • In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow (1) • With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk (1) • Little stored ice remains to melt in the dry season so flow falls very low (1)	3
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Water shortages for irrigation / drinking / HEP in the dry season (1) • Flooding in the wet season (1) • Sudden floods from ice-lake outbursts (glacial lake outburst floods) (1)	2
07.4	<i>1 mark for each point.</i> • Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster; if this exceeds the faster melting at the front, the front advances (1) • But if snowfall at the source has not increased, total volume falls; both retreating and extending glaciers can indicate warming (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
08.1	<i>1 mark.</i> • As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures) (1)	1
08.2	<i>1 mark for the correct answer.</i> • About 39 °C (accept 38-40) (1)	1
08.3	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) (1) • = 1.2 mg dm ⁻³ (accept 1.1-1.3) (1)	2
08.4	<i>1 mark for each point.</i> • Figure 1 shows the maximum / saturation values; processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line (1) • Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster; values above the line are impossible (1)	2
08.5	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Increased metabolic rate so more energy used / less surplus for growth (1) • Eggs develop faster / hatch earlier (out of step with food supply) (1) • Non-indigenous warm-water species may thrive and out-compete indigenous species (1) • Toxicity of other pollutants is greater as toxins are metabolised faster (1) • Reduced resistance to disease (1) • Proteins / enzymes denatured if temperature is very high (above about 40 °C) (1)	3
08.6	<i>1 mark.</i> • Hot water is sprayed from a shower floor in a tall hollow tower; air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	
09.1	<i>1 mark for each point.</i> • 1 nitrogen fixation; 2 nitrification (1) • 3 denitrification; 4 ammonification (1) • 5 ionisation; 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) (1)	3
09.2	<i>1 mark for each point.</i> • Drainage makes the soil aerobic (1) • Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate (1) • Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas (1)	3
09.3	<i>1 mark for each point.</i> • Nitrogen gas is very unreactive so most organisms cannot use it directly (1) • Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate; the cycle recycles a small usable amount (1)	2

Question	Marking guidance	Marks
09.4	<i>1 mark for each point.</i> • Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins (1) • When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: Lasky's principle - vast quantities exist at low grade; reserves vs resource; COOG affected by price, technology, energy costs • Improved exploration: higher resolution multispectral satellite sensors, IR spectroscopy, gravimetry, magnetometry, seismic, resistivity, portable field analysis, targeted trial drilling • Mechanisation allowing deeper underground mines and larger open-cast mines • Low-energy extraction of low-grade ores: bioleaching with acidophilic bacteria and fungi, phytomining, iron displacement, bacterial adsorption of rare earths, polymer adsorption from seawater • Previously inaccessible deposits: polymetallic nodules, deep sea sediments, Antarctica (currently protected) - environmental and legal issues • Extending use: recycling (advantages, difficulties), Cradle to Cradle design, product lifetime extension, substitution of scarce materials, circular economy • Evaluation: environmental impacts vs benefits; energy costs; economics; which strategies are most sustainable; conclusion	9
10.2	<i>1 mark.</i> • All of the material that exists in the lithosphere, including that which will never be technologically or economically exploitable (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147; AQA spec 3.2.3.	

Question	Marking guidance	Marks
11.1	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25

Question	Marking guidance	Marks
11.2	Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Historical data limitations; proxy data - ice cores, isotope ratios (O-18:O-16, C-14), Mg:Ca in calcite, tree rings, pollen, coral • Satellite sensors: visible light imagery, infrared for vegetation and temperature, radar microwaves for wind, wave height, sea level and oil, weather monitoring; passing through cloud • Sensor carriers: ROVs / drones, AUVs, balloons, aircraft, animals; data recorders for abiotic factors; GPS tracking of floats and sampling sites; Argo floats and ocean monitoring • Climate models and their assumptions; monitoring atmospheric composition (Mauna Loa), ozone (Dobson units), sea level and ice • Applications to hydrosphere, minerals (remote sensing surveys), soils, pollution monitoring • Evaluation: improved coverage and resolution vs cost, calibration, interpretation, remaining uncertainty; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 9, 89-107, 402-407; AQA spec 3.2, 3.7.	25